

Final Project + Video Link

January 30, 2026

0.1 VIDEO LINK BELOW

- <https://play.library.utoronto.ca/watch/161e0bd728544446c78e88a89307a42c>
- Please open the link in a new tab
- Please note: Whenever the word ‘correlation’ was used, we meant to say correspondence

1 Exploring Body Mass Correspondence with Extinction Risk Among Reptiles

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1.1.1 Tutorial: TUT0103

1.1.2 TA: Luna Taguchi

1.1.3 Introduction

- How does a reptile’s biological attributes correspond to extinction risk?
- Question: Is there a discernible difference in the average body mass of higher risk and lower risk reptiles?
- 2 Datasets: ‘Amniote_Database_Aug_2015.csv’, ‘IUCN_animals.csv’
- Bootstrapping

```
[7]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
pd.set_option('mode.chained_assignment', None)

amniote_raw = pd.read_csv('Amniote_Database_Aug_2015.csv')
iucn_raw = pd.read_csv('IUCN_animals.csv')
```

```
[8]: display(amniote_raw.head())
display(iucn_raw.head())
```

	class	order	family	genus	species	subspecies	\
0	Aves	Accipitriformes	Accipitridae	Accipiter	albogularis	NaN	
1	Aves	Accipitriformes	Accipitridae	Accipiter	badius	NaN	
2	Aves	Accipitriformes	Accipitridae	Accipiter	bicolor	NaN	
3	Aves	Accipitriformes	Accipitridae	Accipiter	brachyurus	NaN	
4	Aves	Accipitriformes	Accipitridae	Accipiter	brevipes	NaN	

	common_name	female_maturity_d	litter_or_clutch_size_n	\
0	Pied Goshawk	NaN	NaN	
1	Shikra	363.468	3.25	
2	Bicolored Hawk	NaN	2.70	
3	New Britain Sparrowhawk	NaN	NaN	
4	Levant Sparrowhawk	363.468	4.00	

	litters_or_clutches_per_y	...	egg_length_mm	fledging_mass_g	\
0	NaN	...	NaN	NaN	
1	1.0	...	NaN	NaN	
2	NaN	...	NaN	NaN	
3	NaN	...	NaN	NaN	
4	1.0	...	NaN	NaN	

	adult_svl_cm	male_svl_cm	female_svl_cm	birth_or_hatching_svl_cm	\
0	NaN	NaN	NaN	NaN	
1	30.0	NaN	NaN	NaN	
2	39.5	NaN	NaN	NaN	
3	NaN	NaN	NaN	NaN	
4	33.5	NaN	NaN	NaN	

	female_svl_at_maturity_cm	female_body_mass_at_maturity_g	no_sex_svl_cm	\
0	NaN	NaN	NaN	
1	NaN	NaN	NaN	
2	NaN	NaN	NaN	
3	NaN	NaN	NaN	
4	NaN	NaN	NaN	

	no_sex_maturity_d
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

[5 rows x 36 columns]

	assessmentId	internalTaxonId	scientificName	kingdomName	\
0	495630	10030	Hexanchus griseus	ANIMALIA	
1	495907	10041	Heosemys annandalii	ANIMALIA	
2	497499	132523146	Hubbsina turneri	ANIMALIA	
3	497550	10267	Hungerfordia pelewenis	ANIMALIA	
4	498370	10767	Ictalurus australis	ANIMALIA	

	phylumName	orderName	className	familyName	\
0	CHORDATA	HEXANCHIFORMES	CHONDRICHTHYES	HEXANCHIDAE	
1	CHORDATA	TESTUDINES	REPTILIA	GEOEMYDIDAE	

2	CHORDATA	CYPRINODONTIFORMES	ACTINOPTERYGII	GOODEIDAE
3	MOLLUSCA	ARCHITAENIOGLOSSA	GASTROPODA	DIPLOMMATINIDAE
4	CHORDATA	SILURIFORMES	ACTINOPTERYGII	ICTALURIDAE

	genusName	speciesName	infraType	infraName	infraAuthority \
0	Hexanchus	griseus	NaN	NaN	NaN
1	Heosemys	annandalii	NaN	NaN	NaN
2	Hubbsina	turneri	NaN	NaN	NaN
3	Hungerfordia	pelewensis	NaN	NaN	NaN
4	Ictalurus	australis	NaN	NaN	NaN

	authority	redlistCategory \
0	(Bonnaterre, 1788)	Near Threatened
1	(Boulenger in Annandale & Robinson, 1903)	Critically Endangered
2	(de Buen, 1940)	Critically Endangered
3	Beddome, 1889	Endangered
4	(Meek, 1904)	Data Deficient

	redlistCriteria	criteriaVersion	populationTrend	scopes
0	A2bd	3.1	Decreasing	Global
1	A2cd+4cd	3.1	Decreasing	Global
2	B1ab(i,ii,iii,iv)+2ab(i,ii,iii,iv)	3.1	Decreasing	Global
3	B1ab(ii,iii)+2ab(ii,iii)	3.1	Unknown	Global
4	NaN	3.1	Decreasing	Global

2 Methods

- “Amniote_Database_Aug_2015.csv” : ‘class’, ‘genus’, ‘species’, ‘adult_body_mass_g’
- “IUCN_animals.csv” : ‘className’, ‘scientificName’, ‘redlistcategory’
- renamed columns (.rename())
- convert types (convert_dtypes)
- Missing values -> ‘pd.NA’ (.replace())
- Filter the datasets to keep only ‘Reptilia’ (.loc[])

[9]: # Extract necessary columns from the Amniote dataset

```
extract_columns = [
    "class",
    "genus",
    "species",
    "adult_body_mass_g"
]

amniote_data = amniote_raw[extract_columns]

amniote_data.head(18000)
```

```
# Filter out all species that aren't Reptilia
reptilia_data = amniote_data.loc[amniote_data['class'] == 'Reptilia']

reptilia_data
```

```
[9]:
```

	class	genus	species	adult_body_mass_g
14755	Reptilia	Alligator	mississippiensis	62000.0
14756	Reptilia	Alligator	sinensis	14600.0
14757	Reptilia	Caiman	crocodilus	10900.0
14758	Reptilia	Caiman	latirostris	14600.0
14759	Reptilia	Caiman	yacare	17000.0
...	...	...	...	...
21317	Reptilia	Nilssonina	nigricans	6971.0
21318	Reptilia	Palea	steindachneri	NaN
21319	Reptilia	Pelochelys	bibroni	NaN
21320	Reptilia	Pelodiscus	sinensis	2327.5
21321	Reptilia	Trionyx	triunguis	10818.0

[6567 rows x 4 columns]

```
[10]: # Create new column that combines the columns 'genus' and 'species' with ' ' in
      ↪between, used for merging
reptilia_data["scientific_name"] = reptilia_data["genus"] + " " +
      ↪reptilia_data["species"]
reptilia_data.head()

# Rename for a cleaner look
rename_columns = {
    "class": "Class",
    "genus": "Genus",
    "species": "Species",
    "adult_body_mass_g": "Adult Body Mass (g)",
    "scientific_name": "Scientific Name"
}

# Convert data types, replace missing values
amniote_reptilia_data = reptilia_data.rename(columns=rename_columns)
amniote_converted = amniote_reptilia_data.convert_dtypes()
amniote_final = amniote_converted.replace(-999, pd.NA)

amniote_final
```

```
[10]:
```

	Class	Genus	Species	Adult Body Mass (g)	\
14755	Reptilia	Alligator	mississippiensis	62000.0	
14756	Reptilia	Alligator	sinensis	14600.0	
14757	Reptilia	Caiman	crocodilus	10900.0	
14758	Reptilia	Caiman	latirostris	14600.0	

14759	Reptilia	Caiman	yacare	17000.0
...	...	...	...	...
21317	Reptilia	Nilssonina	nigricans	6971.0
21318	Reptilia	Palea	steindachneri	<NA>
21319	Reptilia	Pelochelys	bibroni	<NA>
21320	Reptilia	Pelodiscus	sinensis	2327.5
21321	Reptilia	Trionyx	triunguis	10818.0

	Scientific Name
14755	Alligator mississippiensis
14756	Alligator sinensis
14757	Caiman crocodilus
14758	Caiman latirostris
14759	Caiman yacare
...	...
21317	Nilssonina nigricans
21318	Palea steindachneri
21319	Pelochelys bibroni
21320	Pelodiscus sinensis
21321	Trionyx triunguis

[6567 rows x 5 columns]

```
[11]: # Extract necessary columns from IUCN data set
extract_columns = [
    "className",
    "scientificName",
    "redlistCategory"
]

iucn_data = iucn_raw[extract_columns]

iucn_data.head(20000)

# Filter out all species that are not reptiles
threat_data = iucn_data.loc[iucn_data['className'] == 'REPTILIA']
threat_data
```

```
[11]:
```

	className	scientificName	redlistCategory
1	REPTILIA	Heosemys annandalii	Critically Endangered
7	REPTILIA	Iguanognathus weneri	Data Deficient
9	REPTILIA	Indotestudo forstenii	Critically Endangered
10	REPTILIA	Pangshura sylhetensis	Critically Endangered
28	REPTILIA	Lissemys scutata	Least Concern
...	...	...	...
83388	REPTILIA	Boaedon capensis	Least Concern
83513	REPTILIA	Macroprotodon cucullatus	Least Concern

83580	REPTILIA	Cyclura collei	Critically Endangered
83590	REPTILIA	Chioninia coctei	Extinct
83591	REPTILIA	Leiocephalus eremitus	Extinct

[10148 rows x 3 columns]

```
[12]: # Rename for a cleaner look
threat_rename = {
    "className": "class",
    "scientificName": "scientific name",
    "redlistCategory": "IUCN status",
}

# Convert data types, replace missing values as NA
threat_renamed = threat_data.rename(columns=threat_rename)
threat_converted = threat_renamed.convert_dtypes()
iucn_final = threat_converted.replace(-999, pd.NA)
iucn_final
```

```
[12]:
```

	class	scientific name	IUCN status
1	REPTILIA	Heosemys annandalii	Critically Endangered
7	REPTILIA	Iguanognathus weneri	Data Deficient
9	REPTILIA	Indotestudo forstenii	Critically Endangered
10	REPTILIA	Pangshura sylhetensis	Critically Endangered
28	REPTILIA	Lissemys scutata	Least Concern
...	...	...	...
83388	REPTILIA	Boaedon capensis	Least Concern
83513	REPTILIA	Macroprotodon cucullatus	Least Concern
83580	REPTILIA	Cyclura collei	Critically Endangered
83590	REPTILIA	Chioninia coctei	Extinct
83591	REPTILIA	Leiocephalus eremitus	Extinct

[10148 rows x 3 columns]

- New column in “Amniote_Database_Aug_2015.csv”: ‘scientific_name’
- Facilitates merging
- New column aligns with a column in the other dataset

```
[13]: reptilia_data["scientific_name"] = reptilia_data["genus"] + " " +
      ↪reptilia_data["species"]
reptilia_data.head()
```

```
[13]:
```

	class	genus	species	adult_body_mass_g	\
14755	Reptilia	Alligator	mississippiensis	62000.0	
14756	Reptilia	Alligator	sinensis	14600.0	
14757	Reptilia	Caiman	crocodilus	10900.0	
14758	Reptilia	Caiman	latirostris	14600.0	

14759	Reptilia	Caiman	yacare	17000.0
-------	----------	--------	--------	---------

```

scientific_name
14755 Alligator mississippiensis
14756 Alligator sinensis
14757 Caiman crocodilus
14758 Caiman latirostris
14759 Caiman yacare

```

- Merge datasets (pd.merge())

```

[14]: joined_amniote_iucn_data = pd.merge(
    left=amniote_final,
    right=iucn_final,
    left_on="Scientific Name",
    right_on="scientific name"
)
joined_amniote_iucn_data.head(6020)

```

```

[14]:
   Class      Genus      Species  Adult Body Mass (g) \
0  Reptilia  Alligator  mississippiensis      62000.0
1  Reptilia  Alligator      sinensis      14600.0
2  Reptilia   Caiman   crocodilus      10900.0
3  Reptilia   Caiman  latirostris      14600.0
4  Reptilia   Caiman    yacare      17000.0
...      ...      ...      ...      ...
6015 Reptilia  Nilssonia  nigricans      6971.0
6016 Reptilia   Palea  steindachneri      <NA>
6017 Reptilia  Pelochelys  bibroni      <NA>
6018 Reptilia  Pelodiscus  sinensis      2327.5
6019 Reptilia   Trionyx  triunguis      10818.0

```

```

   Scientific Name      class      scientific name \
0 Alligator mississippiensis REPTILIA Alligator mississippiensis
1 Alligator sinensis REPTILIA Alligator sinensis
2 Caiman crocodilus REPTILIA Caiman crocodilus
3 Caiman latirostris REPTILIA Caiman latirostris
4 Caiman yacare REPTILIA Caiman yacare
...      ...      ...      ...
6015 Nilssonia nigricans REPTILIA Nilssonia nigricans
6016 Palea steindachneri REPTILIA Palea steindachneri
6017 Pelochelys bibroni REPTILIA Pelochelys bibroni
6018 Pelodiscus sinensis REPTILIA Pelodiscus sinensis
6019 Trionyx triunguis REPTILIA Trionyx triunguis

```

```

IUCN status
0 Least Concern
1 Critically Endangered

```

```

2          Least Concern
3          Least Concern
4          Least Concern
...
6015 Critically Endangered
6016 Critically Endangered
6017          Vulnerable
6018          Vulnerable
6019          Vulnerable

```

[6020 rows x 8 columns]

- Exclude 'Data Deficient' reptiles
- Standardize IUCN status names

```

[15]: # Remove reptiles that have insufficient data for IUCN status
filtered_iucn = (joined_amniote_iucn_data['IUCN status'] != 'Data Deficient')

amniote_iucn_final = joined_amniote_iucn_data[filtered_iucn]

# Standardize some IUCN labels
amniote_iucn_final['IUCN status'] = amniote_iucn_final['IUCN status'].
    ↪replace('Lower Risk/near threatened', 'Near Threatened')
amniote_iucn_final['IUCN status'] = amniote_iucn_final['IUCN status'].
    ↪replace('Lower Risk/least concern', 'Least Concern')

# Print final dataframe
amniote_iucn_final

```

```

[15]:
      Class      Genus      Species  Adult Body Mass (g)  \
0    Reptilia  Alligator mississippiensis          62000.0
1    Reptilia  Alligator      sinensis          14600.0
2    Reptilia   Caiman      crocodilus          10900.0
3    Reptilia   Caiman    latirostris          14600.0
4    Reptilia   Caiman        yacare          17000.0
...
6015 Reptilia  Nilssonina      nigricans          6971.0
6016 Reptilia    Palea    steindachneri          <NA>
6017 Reptilia Pelochelys      bibroni          <NA>
6018 Reptilia Pelodiscus      sinensis          2327.5
6019 Reptilia   Trionyx      triunguis          10818.0

      Scientific Name      class      scientific name  \
0    Alligator mississippiensis  REPTILIA  Alligator mississippiensis
1      Alligator sinensis  REPTILIA      Alligator sinensis
2      Caiman crocodilus  REPTILIA      Caiman crocodilus
3      Caiman latirostris  REPTILIA      Caiman latirostris

```


4	Caiman yacare	REPTILIA	Caiman yacare
...	...	...	...
6015	Nilssonina nigricans	REPTILIA	Nilssonina nigricans
6016	Palea steindachneri	REPTILIA	Palea steindachneri
6017	Pelochelys bibroni	REPTILIA	Pelochelys bibroni
6018	Pelodiscus sinensis	REPTILIA	Pelodiscus sinensis
6019	Trionyx triunguis	REPTILIA	Trionyx triunguis

	IUCN status
0	Least Concern
1	Critically Endangered
2	Least Concern
3	Least Concern
4	Least Concern
...	...
6015	Critically Endangered
6016	Critically Endangered
6017	Vulnerable
6018	Vulnerable
6019	Vulnerable

[5503 rows x 8 columns]

```
[16]: # The IUCN status names are very long...let's abbreviate them
rename_status = {
    'Extinct': 'EX',
    'Extinct in the Wild': 'EX in Wild',
    'Critically Endangered': 'CE',
    'Endangered': 'EN',
    'Vulnerable': 'VU',
    'Lower Risk/conservation dependent': 'LR/CD',
    'Near Threatened': 'NT',
    'Least Concern': 'LC'
}

amniote_iucn_final['IUCN status'] = amniote_iucn_final['IUCN status'].
    ↪replace(rename_status)

# Calculate average body mass of species within each category
grouped_data = amniote_iucn_final.groupby('IUCN status', observed=True)
iucn_average_bodymass = grouped_data['Adult Body Mass (g)'].mean()

display(iucn_average_bodymass.sort_values())
```

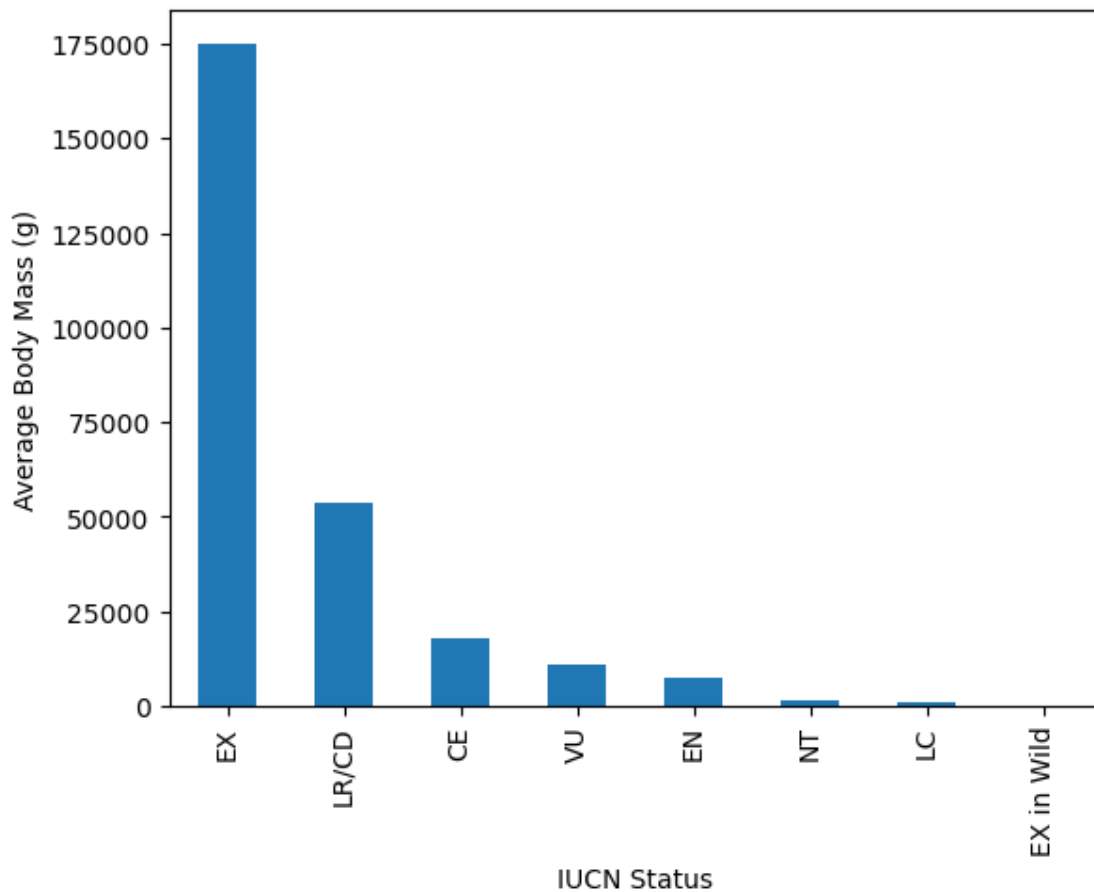
IUCN status	
EX in Wild	2.3
LC	1034.393089
NT	1466.356205

```
EN          7547.857892
VU          10749.619202
CE          17923.774583
LR/CD       53900.0
EX          175000.0
Name: Adult Body Mass (g), dtype: Float64
```

2.0.1 Results

```
[17]: # Graph 1 : Average body mass of each IUCN status category
import matplotlib.pyplot as plt
iucn_average_bodymass.sort_values(ascending=False).plot.bar()
plt.xlabel("IUCN Status")
plt.ylabel("Average Body Mass (g)")
```

```
[17]: Text(0, 0.5, 'Average Body Mass (g)')
```



```
[18]: # Divide the 8 IUCN categories into two: 'Higher Risk' and 'Lower Risk'
# Everything above 'Vulnerable' will be considered 'Higher Risk'
```

```

amniote_iucn_final['IUCN status'] = amniote_iucn_final['IUCN status'].replace({
    'LC' : 'Lower Risk',
    'NT' : 'Lower Risk',
    'LR/CD' : 'Lower Risk',
    'VU' : 'Higher Risk',
    'EN' : 'Higher Risk',
    'CE' : 'Higher Risk',
    'EX in Wild' : 'Higher Risk',
    'EX' : 'Higher Risk'
})

grouped_data = amniote_iucn_final.groupby('IUCN status', observed=True)
iucn_average_bodymass = grouped_data['Adult Body Mass (g)'].mean()

display(iucn_average_bodymass.sort_values())

```

```

IUCN status
Lower Risk      1114.659278
Higher Risk     12464.865743
Name: Adult Body Mass (g), dtype: Float64

```

```

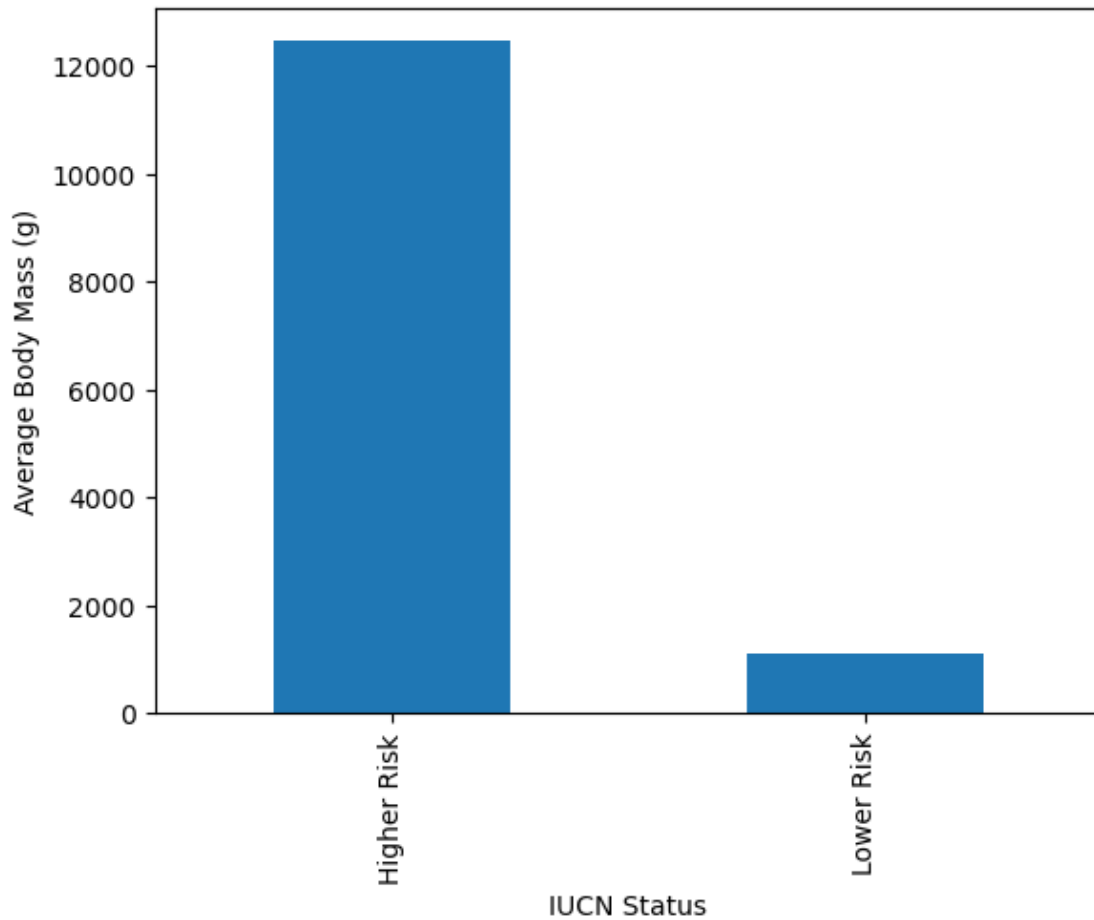
[19]: # Graph 2: Average body mass of higher risk and lower risk species (Higher risk ↪
      ↪= 'Vulnerable' and above)
import matplotlib.pyplot as plt
iucn_average_bodymass.sort_values(ascending=False).plot.bar()
plt.xlabel("IUCN Status")
plt.ylabel("Average Body Mass (g)")

```

```

[19]: Text(0, 0.5, 'Average Body Mass (g)')

```



```
[20]: # Calculate the bootstrapped difference in mean adult body mass between
      ↪ reptiles categorized as "Higher Risk" and "Lower Risk"
      np.random.seed(1009885439)

      lower_risk_data = amniote_iucn_final[amniote_iucn_final['IUCN status'] ==
      ↪ 'Lower Risk']['Adult Body Mass (g)']
      higher_risk_data = amniote_iucn_final[amniote_iucn_final['IUCN status'] ==
      ↪ 'Higher Risk']['Adult Body Mass (g)']

      def boot_single_mean_diff():

          boot_higher_risk = higher_risk_data.sample(frac=1, replace=True)
          boot_lower_risk = lower_risk_data.sample(frac=1, replace=True)

          boot_mass_mean_diff = np.mean(boot_higher_risk) - np.mean(boot_lower_risk)
```

```
    return boot_mass_mean_diff

boot_mass_mean_diff = boot_single_mean_diff()

print(boot_mass_mean_diff)
```

10710.527897288614

```
[15]: # Iterates the code above 5000 times
      np.random.seed(1009885439)

      boot_mean_diff = []

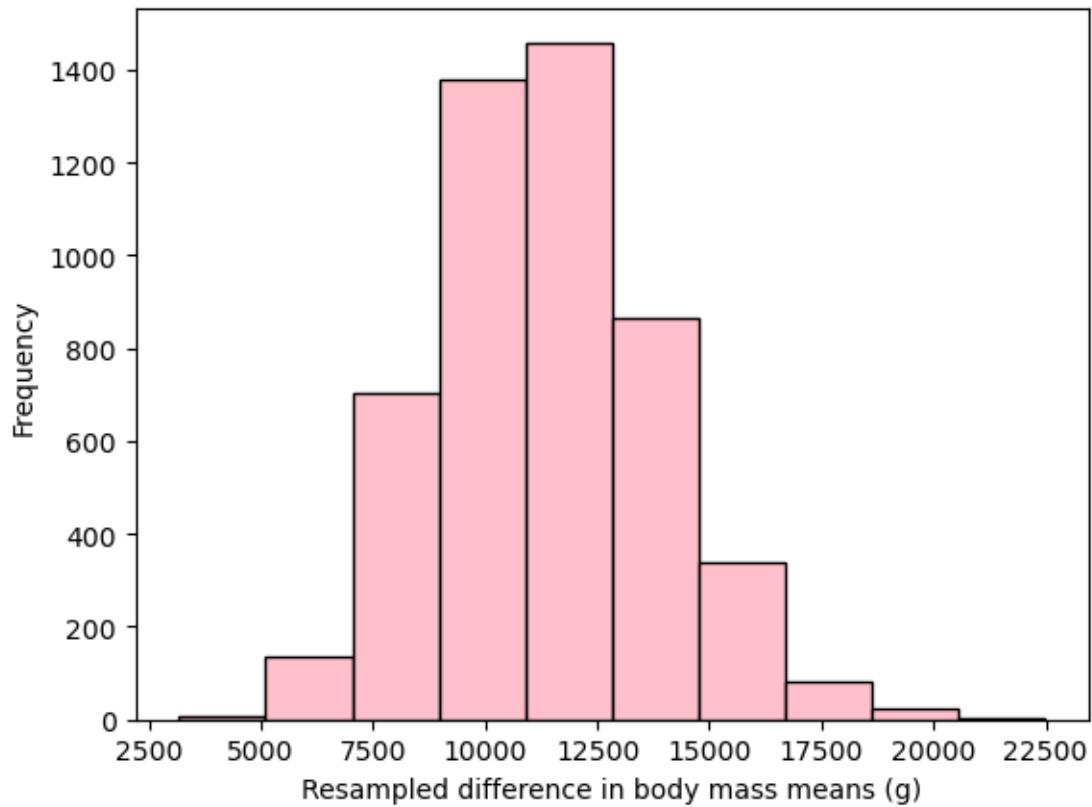
      for _ in range(5000):
          boot_mean_diff.append(boot_single_mean_diff())

      len(boot_mean_diff)
```

[15]: 5000

```
[16]: # Graph 3: Bootstrapped difference in body mass means (Higher risk - Lower risk)
      plt.hist(boot_mean_diff, color="pink", edgecolor='black')
      plt.ylabel("Frequency")
      plt.xlabel("Resampled difference in body mass means (g)")
```

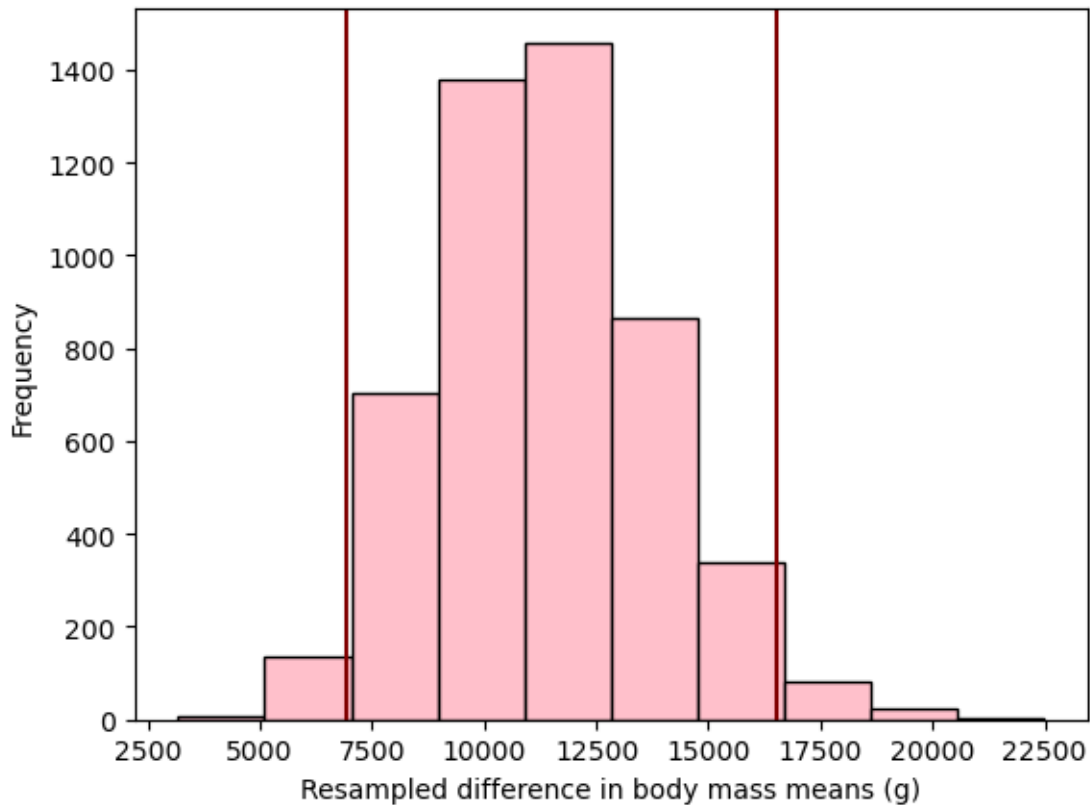
```
[16]: Text(0.5, 0, 'Resampled difference in body mass means (g)')
```



```
[17]: np.percentile(boot_mean_diff, 2.5), np.percentile(boot_mean_diff, 97.5)
```

```
[17]: (6924.266874141668, 16516.658502574723)
```

```
[18]: plt.hist(boot_mean_diff, color="pink", edgecolor="black")
plt.axvline(x=np.percentile(boot_mean_diff, 2.5), color="maroon")
plt.axvline(x=np.percentile(boot_mean_diff, 97.5), color="maroon")
plt.ylabel("Frequency")
plt.xlabel("Resampled difference in body mass means (g));
```



2.0.2 Conclusion

- How does a reptile's biological attributes correspond to their extinction risk?
- Is there a difference in IUCN category between the class Reptilia (reptiles) with larger mean adult body mass and reptiles with smaller mean adult body mass?
- Hypothesis: Compared to their smaller counterparts, reptiles with bigger mean adult body weights are more likely to face survival difficulties
- Amniote_Database_Aug_2015.csv and IUCN_animals.csv data files
- Graph #1: Correspondence Between Body Mass and Extinction Risk
- Threatened species have a higher average body mass compared to those that are not threatened
- Graph #2: Difference in IUCN category based on Body Mass
- Species with higher risk statuses have a higher average body mass than species with smaller average body mass
- Graph #3: Reptile Biological Attributes and Extinction Risk- The resampled difference is positive, higher-risk reptiles have a larger body mass on average compared to lower-risk reptiles
- Larger reptile species, represented by higher average body mass values, tend to have a higher extinction risk. Larger reptile species may have a slower reproductive rate, larger habitat

requirements, and higher energy needs, making them more susceptible to threats

- smaller reptiles species may have higher reproductive rates, more adaptable to environmental changes, and may have less specialized habitat requirements, making them more resilient to extinction threats

2.0.3 Errors, Challenges, Strengths

- Results might contradict actual species status' as a number of factors affect species threat level
- Missing values and inconsistencies affect the reliability of the analysis
- Interpreting results requires knowledge of ecological systems
- The use of multiple datasets
- Educating conservation efforts aimed at protecting reptile species

2.0.4 Improvements:

- Enhancing the data quality to rectify errors and missing values in the datasets
- Use of additional data sources